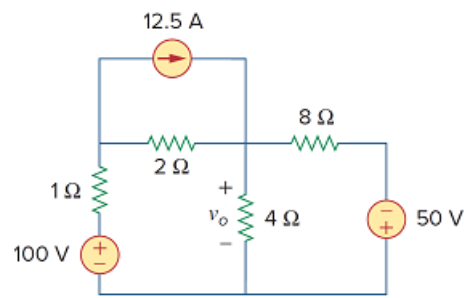


## Problem

Using nodal analysis, find  $v_o$  in the circuit of Fig. 3.63.

**Figure 3.63:**



## Step-by-step solution

## Step 1 of 4

Draw the circuit diagram shown in Figure 1.

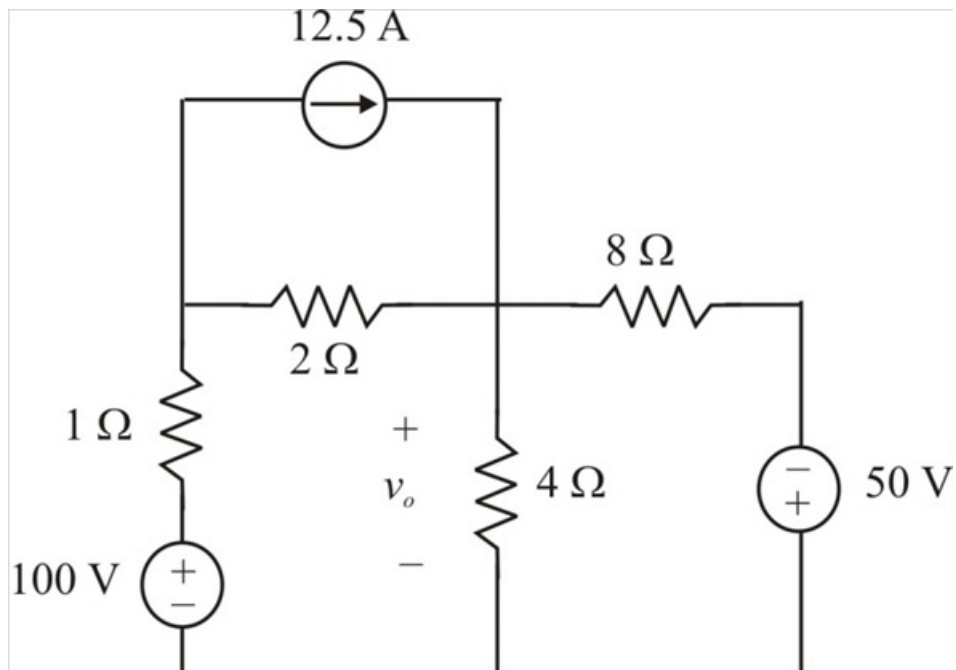


Figure 1

## Step 2 of 4

The circuit is labeled as shown in Figure 2.

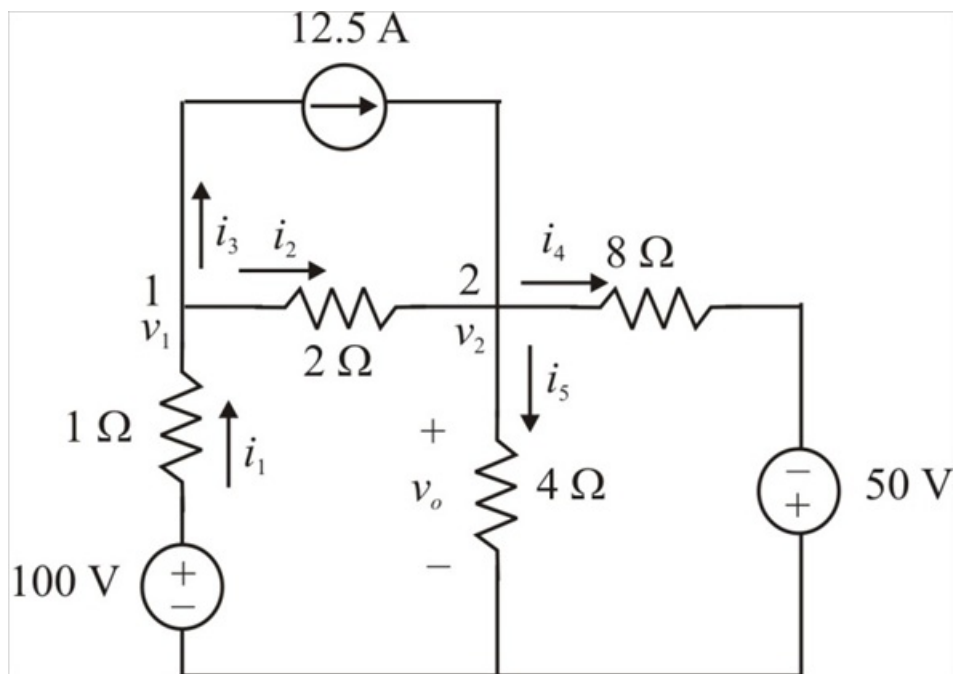


Figure 2

**Step 3** of 4

Applying Kirchhoff's current law at node 1, we get

$$i_1 = i_2 + i_3$$

$$\frac{100 - v_1}{1} = \frac{v_1 - v_2}{2} + 12.5$$

$$200 - 2v_1 = v_1 - v_2 + 25$$

$$3v_1 - v_2 = 175$$

Therefore  $3v_1 - v_2 = 175$  ..... (1)

Applying Kirchhoff's current law at node 2, we get

$$i_2 + i_3 = i_4 + i_5$$

$$\frac{v_1 - v_2}{2} + 12.5 = \frac{v_2 + 50}{8} + \frac{v_2 - 0}{4}$$

$$4v_1 - 4v_2 + 100 = v_2 + 50 + 2v_2$$

$$4v_1 - 7v_2 = -50$$

Therefore  $4v_1 - 7v_2 = -50$  ..... (2)

**Step 4** of 4

Multiplying equation (1) by 7 and subtract equation (2) from it, we get

$$21v_1 - 7v_2 = 1225$$

$$4v_1 - 7v_2 = -50$$

$$17v_1 = 1275$$

$$v_1 = 75 \text{ V}$$

Substituting  $v_1 = 75 \text{ V}$  in equation (1), we get

$$3v_1 - v_2 = 175$$

$$3(75) - v_2 = 175$$

$$v_2 = 225 - 175$$

$$= 50 \text{ V}$$

But

$$v_o = v_2$$

$$= 50 \text{ V}$$

Therefore the node voltage  $v_2$  is 50 V.